

Problema 2 Fie ecuația $f(x) = 0$, $f : [a, b] \rightarrow \mathbb{R}$, $f \in C^3[a, b]$ și α o rădăcină simplă a ei.

(a) Să se arate că

$$x_{k+1} = x_k - \frac{f(x_k)}{\sqrt{f'(x_k)^2 - f(x_k)f''(x_k)}}$$

generează un șir care converge cubic.

(b) Scrieți o funcție care implementează metoda în MATLAB.

(c) Rezolvați ecuația în $\mathbb{R}$ $xe^x - 1 = 0$, folosind funcția de la punctul (b).

Rezolvare

Primul subpunct

Considerăm $f \rightarrow [0, \infty]$, altfel problema n-ar avea sens pe numere reale.

Fie $\phi(x) = x - \frac{f(x)}{\sqrt{f'(x)^2 - f(x)f''(x)}}$. Dorim să găsim rădăcina $f(x_0) = 0$ a funcției f , deci $\phi(x_0) = x_0$. Cautăm ordinul de convergență al acestei funcții:

```
syms f(x) x0;
assume(f(x)>0);
phi(x)=x - f/sqrt(diff(f,x,1)^2 - f*diff(f,x,2))
```

phi(x) =

$$x - \frac{f(x)}{\sqrt{\left(\frac{\partial}{\partial x} f(x)\right)^2 - f(x) \frac{\partial^2}{\partial x^2} f(x)}}$$

```
function phi=substituteWithX0(phi, n, f, x, x0)
    syms h(x);
    for i=1:n
        phi=subs(phi,diff(f,x,i),diff(h,x,i));
    end
    phi=subs(phi,f(x),0);
    phi=subs(phi,h,f);
    phi=subs(phi,x,x0);
    phi=simplify(phi);
end
```

```
phi_df1=diff(phi, x, 1)
```

phi_df1(x) =

$$\frac{f(x) \left(\frac{\partial^2}{\partial x^2} f(x) \frac{\partial}{\partial x} f(x) - f(x) \frac{\partial^3}{\partial x^3} f(x) \right)}{\sigma_1^{3/2} 2} - \frac{\frac{\partial}{\partial x} f(x)}{\sqrt{\sigma_1}} + 1$$

where

$$\sigma_1 = \left(\frac{\partial}{\partial x} f(x) \right)^2 - f(x) \frac{\partial^2}{\partial x^2} f(x)$$

phi_df1(x0)=substituteWithX0(phi_df1,2,f,x,x0)

phi_df1(x0) =

$$1 - \frac{\frac{\partial}{\partial x_0} f(x_0)}{\sqrt{\left(\frac{\partial}{\partial x_0} f(x_0) \right)^2}}$$

Deoarece derivata este pozitiva (?), $\Rightarrow \phi'(x_0) = 0$.

phi_df2=diff(phi, x, 2)

phi_df2(x) =

$$\frac{\sigma_2 \frac{\partial}{\partial x} f(x)}{\sigma_1^{3/2}} - \frac{f(x) \left(f(x) \frac{\partial^4}{\partial x^4} f(x) - \left(\frac{\partial^2}{\partial x^2} f(x) \right)^2 \right)}{\sigma_1^{3/2} 2} - \frac{f(x) \sigma_2^2 3}{\sigma_1^{5/2} 4} - \frac{\frac{\partial^2}{\partial x^2} f(x)}{\sqrt{\sigma_1}}$$

where

$$\sigma_1 = \left(\frac{\partial}{\partial x} f(x) \right)^2 - f(x) \frac{\partial^2}{\partial x^2} f(x)$$

$$\sigma_2 = \frac{\partial^2}{\partial x^2} f(x) \frac{\partial}{\partial x} f(x) - f(x) \frac{\partial^3}{\partial x^3} f(x)$$

phi_df2(x0)=substituteWithX0(phi_df2,4,f,x,x0)

phi_df2(x0) = 0

Deci si $\phi''(x_0) = 0$

phi_df3=diff(phi,x,3)

phi_df3(x) =

$$\frac{3 \sigma_1 \sigma_7}{\sigma_3} - \frac{3 \sigma_2 \frac{\partial}{\partial x} f(x)}{\sigma_3} - \frac{f(x) \left(\sigma_5 \frac{\partial}{\partial x} f(x) - 2 \sigma_4 \sigma_7 + f(x) \frac{\partial^5}{\partial x^5} f(x) \right)}{\sigma_6^{3/2} 2} - \frac{9 \sigma_1^2 \frac{\partial}{\partial x} f(x)}{4 \sigma_6^{5/2}} - \frac{\sigma_4}{\sqrt{\sigma_6}} + \frac{f(x) \sigma_1^3}{\sigma_6^{7/2} 8}$$

where

$$\sigma_1 = \sigma_7 \frac{\partial}{\partial x} f(x) - f(x) \sigma_4$$

$$\sigma_2 = f(x) \sigma_5 - \sigma_7^2$$

$$\sigma_3 = 2 \sigma_6^{3/2}$$

$$\sigma_4 = \frac{\partial^3}{\partial x^3} f(x)$$

$$\sigma_5 = \frac{\partial^4}{\partial x^4} f(x)$$

$$\sigma_6 = \left(\frac{\partial}{\partial x} f(x) \right)^2 - f(x) \sigma_7$$

$$\sigma_7 = \frac{\partial^2}{\partial x^2} f(x)$$

```
phi_df3(x0)=substituteWithX0(phi_df3,5,f,x,x0)
```

```
phi_df3(x0) =
```

$$\frac{\left(3 \left(\frac{\partial^2}{\partial x_0^2} f(x_0) \right)^2 - 4 \frac{\partial^3}{\partial x_0^3} f(x_0) \frac{\partial}{\partial x_0} f(x_0) \right) \left(\frac{\partial}{\partial x_0} f(x_0) \right)^3}{4 \left(\left(\frac{\partial}{\partial x_0} f(x_0) \right)^2 \right)^{5/2}}$$

Observam ca aici nu se mai reduc argumentele, deci $\phi'''(x_0) \neq 0$.

$\phi'(x_0) = \phi''(x_0) = 0, \phi'''(x_0) \neq 0 \Rightarrow$ ordinul de convergenta al sirului x_n este 3 $\Rightarrow$ sirul converge cubic.

Al doilea subpunct

```
function [x1,val_f,i]=NewtonV2(f,df,ddf,x0,err,NrMaxIt)
    if nargin < 6, NrMaxIt=100; end
    if nargin < 5, err=1e-12; end

    for i=1:NrMaxIt
        x1=x0-f(x0)/sqrt(df(x0)^2 - f(x0)*ddf(x0));
```

```

        if norm(f(x1),Inf)<err||norm(x1-x0,Inf)<err...
            ||norm(x1-x0,Inf)/norm(x1,Inf)<err
            val_f=f(x1);
            return;
        end
        x0=x1;
    end
    val_f=f(x1);
end

```

Al treilea subpunct

```

syms x;
f=x*exp(x)-1

```

$$f = x e^x - 1$$

```
df=diff(f,x,1)
```

$$df = e^x + x e^x$$

```
ddf=diff(f,x,2)
```

$$ddf = 2 e^x + x e^x$$

```

f=matlabFunction(f);
df=matlabFunction(df);
ddf=matlabFunction(ddf);

```

```
[x, val_f, nr_it]=NewtonV2(f,df,ddf,0)
```

```

x =
    0.567143290409784
val_f =
    0
nr_it =
    3

```

```
c0=fzero(f, 0)
```

```

c0 =
    0.567143290409784

```